

External Visual Memory Reranking for Frozen Game World Models

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Abstract

001 Long-horizon world generation often drifts: objects dis-
002 appear, spatial cues move, and the model forgets first-
003 visit evidence needed for later consistency. This pa-
004 per studies memory modules for game world models
005 through the AP0006 Stage A/Stage C structure. Stage
006 A is reserved for a teammate’s forthcoming analysis
007 of MatrixGame 3.0. Our implemented contribution is
008 Stage C, a lightweight external memory module built
009 on top of frozen MatrixGame-2 at inference time. The
010 module writes first-visit visual evidence as object crops
011 and descriptors, reads memory through patch-level fea-
012 ture matching between stored crops and sampled can-
013 didate frames, and reranks generated candidate videos.
014 With approved road-sign memory, the selected output
015 changes from seed1 to seed8. The resulting mecha-
016 nism is lightweight, interpretable, and does not modify
017 model weights, attention, or training.

018 1. Introduction

019 World models can generate visually plausible short
020 clips, yet long-horizon consistency remains difficult. A
021 scene may initially contain a useful landmark or tar-
022 get object, but later generations can forget where it
023 was, replace it with distractors, or select a rollout that
024 no longer respects earlier evidence. This motivates a
025 practical memory question for generative world mod-
026 els: what should be stored, when should memory be
027 read, and how should it influence later outputs?

028 Our AP0006 project follows a two-stage structure.
029 Stage A is reserved for teammate analysis of Ma-
030 trixGame 3.0 memory behavior. Our main imple-
031 mented contribution is Stage C on frozen MatrixGame-
032 2, where we add an inference-time external memory
033 module without modifying the base generator. The
034 module stores first-visit visual evidence, compares it
035 against regular patches sampled from candidate videos,
036 and reranks the candidate pool using an interpretable
037 similarity score. The goal is not to improve raw real-

ism, but to study whether a lightweight memory inter-
face can change which generated rollout is selected for
better object-level consistency.

2. Stage A: MatrixGame 3.0 Placeholder

TODO_TEAMMATE_MATRIXGAME3_STAGE_A
Team placeholder. This section is reserved for the
teammate responsible for analyzing MatrixGame 3.0.
The final version should identify where MatrixGame
3.0 writes, reads, and updates memory, what data
structures represent memory, which code locations im-
plement the mechanism, and how one concrete input
example demonstrates memory influence. We inten-
tionally leave this section compact in the current draft
to avoid inventing unverified details.

3. Stage C: External Visual Memory Reranking

3.1. Write and store

Stage C targets frozen MatrixGame-2 with the gener-
ator kept unchanged throughout the pipeline. We im-
plement memory entirely outside the model and apply
it as external inference-time memory. When the first
visit to a scene reveals a task-relevant object, such as a
road sign, we write that evidence into a small memory
bank as an object crop together with its feature vec-
tor and lightweight metadata. Typical fields include
the crop path, frame index, candidate or scene iden-
tifier, and a short semantic tag indicating the target
object. This design keeps the stored unit explicit and
auditable: the memory bank contains visual evidence
rather than hidden recurrent state. Figure 1 summa-
rizes the full write-read-rerank pipeline.

3.2. Read, match, and score

At read time, the reranker evaluates a candidate set
of generated videos indexed by j . From candidate j ,
we sample frames indexed by t and split each frame
into regular grid patches indexed by k . Let m denote a
stored memory crop, $p_{j,t,k}$ the k -th patch from frame t

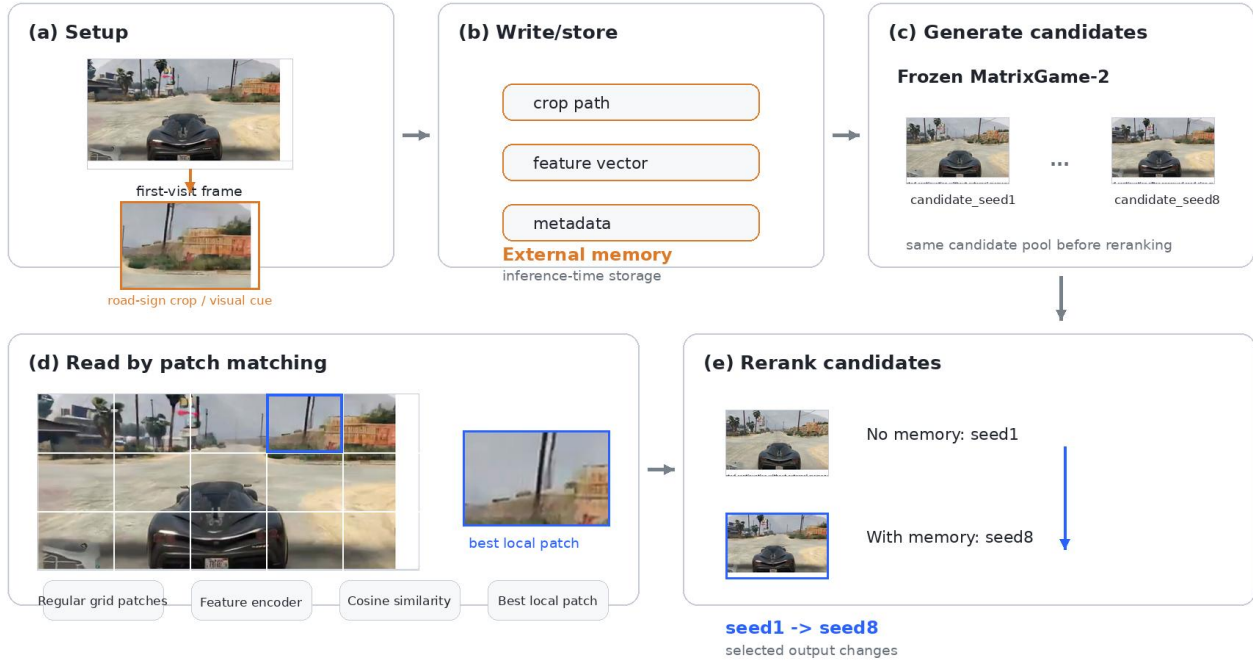


Figure 1. Overview of the external memory reranking pipeline. MatrixGame-2 is kept frozen. The memory module writes first-visit visual evidence, reads it through regular-grid patch matching, and reranks generated candidate videos. The main observed selection changes from seed1 without memory to seed8 with approved visual memory.

of candidate j , and $f(\cdot)$ a visual feature encoder. Depending on environment availability, f can be instantiated with DINOv2, CLIP, torchvision ResNet18, or a normalized RGB fallback. These components are used as feature extractors only; they are not segmentation models.

After feature normalization, we score each patch against memory using cosine-style similarity:

$$s_{j,t,k} = \cos(f(m), f(p_{j,t,k})). \quad (1)$$

For each frame, we keep the strongest matching patches and average them:

$$S_{j,t} = \frac{1}{K'} \sum_{k \in \text{TopK}} s_{j,t,k}. \quad (2)$$

Frame-level scores are then averaged across sampled frames to obtain a candidate-level memory score,

$$S_j = \frac{1}{T} \sum_t S_{j,t}, \quad j^* = \arg \max_j S_j. \quad (3)$$

The reranker finally selects the candidate with the largest S_j . Figure 2 illustrates this patch-level memory reading process with a stored crop, regular grid

patches, and the selected best-matching cell. In words, memory is used neither to retrain the generator nor to alter its internal attention, but to select the roll-out whose visible patches best agree with first-visit evidence.

4. Experiments and Results

Our current evidence is a compact case study centered on road-sign memory. The baseline system samples a pool of candidate videos from frozen MatrixGame-2 and selects a default candidate without external memory. The Stage C reranker reuses the same candidate pool but changes the final choice according to object-patch similarity. In the approved road-sign setting, the selected candidate changes from seed1 to seed8, showing that external memory can alter output selection even when the generator itself is unchanged.

The accompanying project artifacts provide qualitative support for this switch: a generated comparison demo and the patch-level illustration in Figure 2. We treat these assets as evidence for candidate reranking, not as proof that MatrixGame-2 has acquired an internal memory mechanism. The contribution here is the external reranker itself: a simple memory card, a trans-

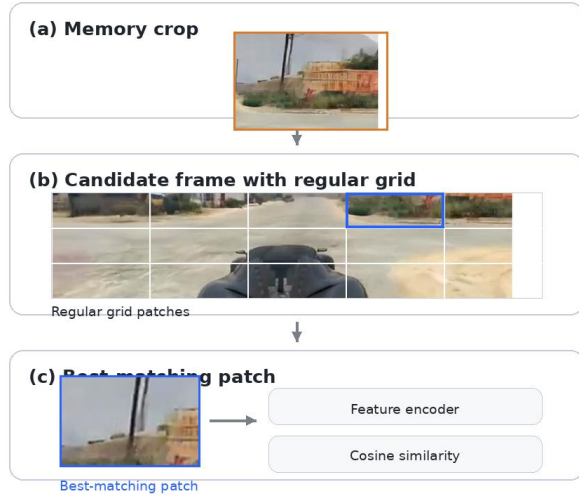


Figure 2. Patch-level memory read. A stored memory crop is compared with regular-grid patches from candidate frames using a visual feature encoder and cosine-style similarity. No object segmentation is used; patches are regular grid cells.

Setting	Memory	Selected Seed	Interpretation
No memory	none	seed1	default choice
Road-sign memory	crop + patch match	seed8	selection switches

Table 1. External memory changes candidate selection in the MatrixGame-2 candidate pool.

generation quality, but an explicit memory mechanism for consistency-oriented candidate selection.

References

parent matching rule, and an observable difference in which candidate is chosen.

5. Limitations and Conclusion

This study has several limitations. First, the mechanism is external reranking only; it does not modify model parameters, hidden state, or attention inside MatrixGame-2. Second, the current evidence is limited to a small candidate pool and a small number of scenarios, so the results should be interpreted as qualitative or small-scale case-study evidence rather than broad statistical validation. Third, Stage A remains pending teammate contribution, so the full paper will need a verified MatrixGame 3.0 analysis before final submission.

Even with these constraints, the project clarifies a useful design point for memory modules in world models. A lightweight bank of first-visit visual evidence, combined with patch-level matching and candidate reranking, can produce an interpretable change in selected output from seed1 to seed8 without retraining the base model. The contribution is therefore not raw